

Failure-Tolerant and Power-Optimal ADCS design for 3U CubeSats

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Abstract

Nanosatellites have emerged over the past decade as a cost-effective solution for low Earth orbit missions that do not require large payload volumes. Among them, CubeSats have gained relevance, especially in academic and student-led projects. Their design, however, remains highly challenging due to stringent constraints on volume, power consumption, and attitude accuracy imposed by mission requirements and payload specifications. Typical CubeSat missions involve operation modes, including Nadir-pointing, Sun-pointing and Point-Tracking, answering needs for data transmission, solar charging and ground station tracking. These modes demand an actuator configuration capable of versatile control (typically combining magnetic torquers and reaction wheels) and a robust sensor layout for attitude determination. Consequently, the design of the Attitude Determination and Control System is critical, requiring not only precise control and estimation capabilities but also resilience to actuator or sensor failures, all under tight power and computational constraints. In this work, a failure-tolerant ADCS architecture for 3U CubeSats is proposed, integrating redundant actuators and sensors, followed by a comparative study on power-efficient control. This study aims to explore ways to balance reliability with energy efficiency, mitigating the additional power demand introduced by redundancy through optimized control allocation. Computational efficiency is also a key objective, ensuring onboard feasibility. Several control approaches are investigated through a developed realistic MATLAB simulation, including classical Proportional-Integral-Derivative control, Linear Quadratic Regulation in both finite and infinite horizon formulations, and Model Predictive Control. On the estimation side, sensor fusion techniques for accurate attitude reconstruction under disturbances are employed, exploring the performance of a Multiplicative Extended Kalman Filter subject to model mismatch. This study contributes a systematic comparison of control strategies tailored for power- and computation-limited CubeSat platforms, emphasizing practical implementability of fault-tolerant ADCS architectures in future 3U missions.

Keywords

ADCS, CubeSat, Digital Twin

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Nomenclature

$\mathbf{h}_b$	Satellite's body angular momentum vector
ϕ	Reaction Wheel tilt angle
$\mathcal{F}_o$	Orbit Frame
q	Attitude quaternion
$\boldsymbol{\omega}$	Angular speed vector
$\hat{A}, \hat{B}, \hat{C}$	Linearized state-space matrices
$\mathbf{y}$	System output vector
$\mathbf{m}_b$	Magnetic moment vector

Note that vectors are denoted in bold.

Acronyms/Abbreviations

ADCS	Attitude Determination and Control System
CSS	Coarse Sun Sensor
DGM	Direct Geometric Model
EHS	Earth Horizon Sensor
FBLin	Feedback Linearization
FH	Finite Horizon
FSS	Fine Sun Sensor
GNSS	Global Navigation Satellite System
IGM	Inverse Geometric Model
IH	Infinite Horizon
LQR	Linear Quadratic Regulator
MEKF	Multiplicative Extended Kalman Filter
MPC	Model Predictive Control
MTM	Magnetometer
MTQ	Magnetic Torquer
PP	Pole-Placement
RW	Reaction Wheel
S&A	Sensor and Actuator
TRIAD	TRIaxial Attitude Determination

1. Introduction

The Attitude Determination and Control System (ADCS) is a crucial part of satellite architecture. Other subsystems rely heavily on its capabilities, like telecommunication, the electrical power system or payloads. Redundancy needs naturally follow, to accommodate Sensors and Actuators (S&A) failures. Consequently, constraints associated with volume, power and complexity arise, highlighting that this kind of design is based on trade-offs between safety and practicality. Small

satellites design, including 3-Units CubeSat design, is typically subjected to these restrictions, and will be in the center of this work. Their application has been growing over the past decades, whether through student projects or simply through low-volume payloads missions. Typical in-operation modes count Sun pointing for charging, Nadir pointing for isotropic telecommunication and (Earth) point tracking for precise telecommunication or payload scientific measurements. For computation reasons, control schemes employed on-board usually use low-level designs, even though more computationally heavy ones may provide valuable energy savings. This leads to the subject presented in this work, where the objective is not to provide a completely fail-proof system, but rather a sweet spot between complexity and robustness to failure.

The work presented in this paper is divided into two main parts. First, an S&A layout suited for the aforementioned modes is explored, where associated design choices regarding their failure tolerance and impact on in-operation performances are highlighted. Second, a set of controllers is compared with regards to their computation load, power-efficiency and pointing performances, using a specifically developed digital twin of a 3-Units CubeSat in MATLAB.

2. ADCS Architecture

In this section, attention is brought to the choice of S&A for a typical 3-Units CubeSat mission, and includes insight on sizing, placement and other specific pieces of advice.

2.1. Actuators

A proper set of actuators is essential for precise and effective attitude control. Recently, control momentum gyroscopes showed appealing performances, especially for their high torque to-volume ratio. However, focus here is made on their older counterparts, Reaction Wheels (RWs) notably for their extensive flight heritage through missions of all sizes. While 3 of them would theoretically suffice to ensure a 3-axis (roll, pitch yaw) control authority, a failure of a single one of them would lead to uncontrollability, and inevitably loss of the mission. For this reason, a set of 4 RWs is preferred, adding redundancy while keeping a reasonable amount of volume within the satellite's structure. Placement with regards to the satellite's center of mass can greatly affect jitter, and ought to be verified for jitter sensitive payloads. Despite various arrangements

existing for 4 RWs [1], a low-volume symmetrical arrangement like the pyramidal one stands out for its practicality upon failure, since a single 3-RWs model can be derived prior to launch.

A simple Direct Geometric Model (DGM), written K_{DGM} , relates body angular momenta to the RWs momenta like in Eq. 1.1.

$$\mathbf{h}_b = K_{DGM} \mathbf{h}_{RW} \quad (1.1)$$

$$\mathbf{h}_{RW} = K_{IGM} \mathbf{h}_b \quad (1.2)$$

Because we have $K_{DGM} \in \mathbb{R}^{3 \times 4}$, its associated Inverse Geometric Model (IGM), namely K_{IGM} , can be defined as its Moore-Penrose pseudo-inverse as in Eq. 1.2. The failed-DGM is simply the reduced vector spaces spanned by the active RW columns within K_{DGM} . An offset vector representing a still spinning but uncontrollable wheel can be added to form the relationship presented in Eq. 2.

$$\mathbf{h}_b = K_{DGM}^{active} \mathbf{h}_{RW}^{active} + K_{DGM} \mathbf{e}_i \mathbf{h}_{RW}^i \quad (2)$$

Where $\mathbf{e}_i$ is the index vector associated with failed RW i .

Naturally, the tilt angle ϕ of the arrangement greatly affects its capabilities and can be optimized [2] based on the spacecraft's moment of inertia among others. A naïve approach can nonetheless offer close-to-optimal performance and is actually used by some CubeSat hardware-oriented companies.

Another crucial aspect regarding RWs is the angular momentum envelope (or equivalently, the angular speed envelope), used mainly to conservatively compute slew rate. A polytope can be defined as in Eq. 3, from which the slew rate can be obtained, represented by the center of largest inscribed ball within the convex hull associated with $\mathcal{H}$.

$$\mathcal{H} = \{\mathbf{h}_b \in \mathbb{R}^3 | \mathbf{h}_b = K_{DGM} \mathbf{h}_{RW},$$

$$\|\mathbf{h}_{RW}\|_\infty \leq h_{RW}^{sat}\} \quad (3)$$

This methodology can be applied to a failed system, where the relationship between $\mathbf{h}_b$ and $\mathbf{h}_{RW}$ in Eq. 3 is replaced by the one in Eq. 2. This slew rate computation relates how fast the CubeSat can move, and consequently how far the settling time in-between modes extends, which makes it a key parameter to consider.

The second type of widely used actuator is Magnetic Torquers (MTQs). Their usage is however limited to the orbit characteristics, as they rely on Earth magnetic field direction and magnitude. These actuators are especially useful to perform angular momentum dumping of RWs and extend control authority, allowing

effective attitude control in scenarios where more than one RW would have failed [3].

A set of 3 orthogonal MTQs is typically preferred. Because of their basic solenoid design, and since redundancy cannot be achieved by simply adding another, this amount is considered sufficient. Moreover, the generated magnetic fields can be harmful to other subsystems, thus adding more of them would lead to additional architectural issues. Although, for missions heavily relying on MTQs for attitude control like in [4], two sets of 3 orthogonal MTQs could be implemented.

Attention shall be brought to their placement, as none of their generated magnetic field should interfere with one another. Similarly, their remnant magnetic field should be negligible to ensure magnetic cleanliness. A standard approach is to apply a low duty cycle for MTQs, allowing measurements when no interference can be felt. Regarding MTQ sizing, a B-dot sequence for detumbling considering worst-case tip-off rates should be simulated, as analytical derivations may provide underestimated results.

Using RWs paired with MTQs can offer an ADCS architecture very tolerant to failure, with high control authority and easy control system design.

2.2. Sensors

Sensors provide the satellite with required information for attitude determination. They ought to provide sufficient measurements during different orbit phases, including eclipse. It follows that a set of redundant gyroscopes is necessary to obtain angular speed information at all times. To them we add a redundant set of Magnetometers (MTMs), allowing Earth magnetic field knowledge. These two types of sensors are fundamental, and with a suitable Earth magnetic field model and a reference vector, can provide attitude estimates by themselves. Note that MTMs are very sensitive to magnetic cleanliness on board. Subsystems emitting magnetic fields shall be placed as far as possible to avoid saturation of the measurements. Sometimes, a convenient approach is to apply a duty cycle to emitting subsystems like MTQs, addressed above. A deployable boom is also an option to be considered, especially in hardware-dense spacecrafts.

Nevertheless, supplemental information about the environment is needed, so as to have a reference to track. An on-board Global Navigation Satellite System (GNSS) would typically supply information on the orbit position

and velocity, allowing accurate frame relationships for reference and state transformations. That is, it is crucial in order to calibrate on-board models. However, being a whole subsystem itself, its failure analysis extends out of the scope of this work.

For sun pointing, the reference can be obtained with a Fine Sun Sensor (FSS), which can measure pitch and roll relative to the sun vector with high accuracy. Up to one FSS per face with solar panels can be implemented. Coarse Sun Sensors (CSSs) offer a coarser yet sufficient estimate of the sun vector for sun capture and therefore should be considered in addition to the FSS. Besides, they provide redundancy to the architecture, meaning a failure of the single FSS will reduce sun pointing performances while keeping charging possible. The disposition of CSSs should be such that a field of view coverage close to 4π is attained. This way, sun capture remains possible in almost every attitude scenario apart during eclipse, while the failure of one or several CSSs remains not critical.

For nadir pointing and point-tracking, knowledge about $\mathcal{F}_o$ provided by the GNSS is used. Additionally, a high-accuracy sensor can be added depending on mission needs. An Earth Horizon Sensor (EHS) has the advantage of being usable during eclipse, while a Star Tracker is able to provide a full 3-axis measurement with higher precision. This choice falls back to on-board resources availability and accuracy needs.

Note that even if the aforementioned sensor layout allows a typical mission to perform Sun Pointing, Nadir Pointing and Point-Tracking, sizing of these sensors' accuracies should be made with regards to attitude error tolerance and specified error margins.

3. Simulation

Redundancy of components brings robustness and reliability but also comes with increased power consumption. In order to tackle it, efficient control strategies are investigated.

3.1. Digital Twin

A satellite attitude simulator developed in MATLAB serves as a realistic benchmark. It includes orbit dynamics, attitude kinematics and dynamics using quaternions, as well as S&A modelling. S&As include: a 4 RWs pyramidal architecture with ϕ_{naive} , 3 orthogonal MTQs, 2 MTMs, 2 Gyroscopes, 1 FSS and 1 EHS. A high

duty cycle (0.9) for periods with zero input is modelled, and actuators saturation limits are respected. A 3-Units CubeSat is simulated at 1 Hz, with all solar panels positioned on its Z + face. Orbit altitude, eccentricity and initial attitude are all randomized around nominal values at each start to obtain statistically significant results. An exhaustive list of configuration parameters can be found in the associated GitHub repository [7], as they cannot be all discussed here. The satellite is simulated for one orbit duration, and a reference signal indicating mode changes is defined as follows: orbit frame alignment, sun pointing, nadir pointing, ground station tracking and back to sun pointing. This allows us to test control strategies on different modes, as well as their transition performances, while keeping a realistic scenario.

Nonetheless, several features have not been implemented, counting CSSs and sun capture, S&A failure, actuator reading error, detumbling phase, MTQ remnant magnetic field and disturbance torques (solar radiation, gravity gradient, ...).

Furthermore, different assumptions were made. Orbit position and velocity are considered ideally known, which is reasonable if a GNSS is present on board. The noise model is considered gaussian with zero-mean, which is also reasonable in view of proper calibration prior to launch, and since the external disturbance comes from numerous sources.

Overall, arguments are made to highlight this digital twin implementation offers valuable results with regards to a classic 3-Units CubeSat mission.

3.2. Control and State Estimation schemes

To reject the sensor noise (white noise, random walk), a Multiplicative Extended Kalman Filter (MEKF) [6] is used. It allows quaternion representation in the estimate and uses linearized matrices around the latest state estimate. At first, 3σ sensor measurement bounds used in the noise model are known. Then, a model mismatch is induced to test robustness to model discrepancies. This mismatch is represented by badly tuned process noise and measurement noise covariance matrices, i.e. identity matrices. Estimates are built based on an output vector y obtained by TRIaxial Attitude Determination (TRIAD) [5], when two non-colinear measurement vectors are available. If such two

vectors are not available, the predicted state by the MEKF serves as estimate. This ensures proper fusion of measurements, avoids gimbal lock issues and eliminates mathematical singularities associated with vector alignment. Since the plant we aim to control is nonlinear, several control strategies are possible. Among others, a Proportional-Derivative controller obtained by Feedback Linearization (FBlin) is tested. It reduces dynamics to a simple double integrator, which can be tuned using classical conventions. Linearized controllers, including Pole-Placement (PP), Finite-Horizon Linear Quadratic Regulator (FH-LQR), Infinite-Horizon Linear Quadratic Regulator (IH-LQR) and Model Predictive Control (MPC) are also added for comparison. They make use of a linearized state-space representation as presented in Eq. 4.1 and Eq. 4.2.

$$\frac{d}{dt} \begin{bmatrix} q \\ \omega \end{bmatrix} = \hat{A} \begin{bmatrix} q \\ \omega \end{bmatrix} + \hat{B} h_{rw} \quad (4.1)$$

$$y = \hat{C} \begin{bmatrix} q \\ \omega \end{bmatrix} \quad (4.2)$$

The discrete state-space is obtained by forward Euler discretization to preserve the sparsity structure of the matrices. Notice that the control input considered is only RWs, as the control input for MTQs is defined externally as a proportional feedback controller to passively desaturate RWs. The MPC controller is, however, defined with extended dynamics, considering h_{rw} in the state and m_b in the input vector, offering more control authority, and making actuator constraint handling possible. Notice also that this state includes the attitude and its derivate, meaning the computed gains are no other than a PD controller gains. One can prove both state-space definitions are reachable, and from a complete output obtained by TRIAD, *i.e.* $\hat{C} = I$, it follows that the system is observable. Finally, the FH-LQR controller uses a horizon of 10 timesteps, while the MPC controller uses an 8 timesteps horizon. All hyperparameters, including gains and cost matrices, are hand-tuned for all controllers.

4. Results

The results are presented in Figure 1 and Figure 2 for the 5 considered controllers, first using a well-tuned MEKF and second with a model mismatch. Each set of experiments has been repeated 100 times to obtain statistically representative results.

Using a correctly tuned MEKF, computationally intensive controllers (IH-LQR, MPC) take significantly more time to provide the control law: the MPC controller is 37 times slower than

the FBlin controller. The FBlin controller clearly outperforms the others in terms of power consumption with a median of 2.6 kW compared to 4.4 kW for the FH-LQR controller. All controllers have statistically close performance results, regarding their high variance; the FBlin controller however performs slightly worse (median 1 deg off from the PP controller). The trade-off between performance and power consumption can be highlighted for the MPC controller, which seems to be very consistent in its consumption (first and third quartile being 360W apart), for variable performance results (two times more interquartile range than the FBlin controller).

In the mismatch plots, the FBlin controller shows significantly higher power consumption ($\approx 300\%$). This is due to the way this controller works and shows how sensitive it is to model discrepancies. Linearized controllers on the other hand suffer from a 30% increase in angle error for only a 10% increase in power consumption. In terms of median, the MPC controller appears to have the best angle error to consumption ratio ($2.6 \frac{\text{deg}}{\text{kW}}$), followed by FH-LQR with $2 \frac{\text{deg}}{\text{kW}}$ and others being lower than $1.7 \frac{\text{deg}}{\text{kW}}$. However, the FH-LQR computation time per iteration is almost 25 times lower median-wise, compared to the MPC controller.

5. Discussion

Results obtained through the digital twin showed interesting relationships between robustness, power efficiency and pointing performance. Several points need, however, to be highlighted to understand how results could be biased, and consequently, improved.

First, as mentioned previously, no process noise is induced as no feature regarding actuator reading error has been implemented. This implies that every prediction step of the MEKF is perfectly modelled. Nevertheless, this has been tackled using mismatched noise covariance matrices and should suffice as to express their effect.

Second, controllers have been hand-tuned, and most likely the current results do not speak for their full potential. Machine learning heuristics could be employed to tune them, for example using particle swarm optimization or genetic algorithms. It can be expected that this will close the performance gap between the FBlin controller and optimization-based controllers.

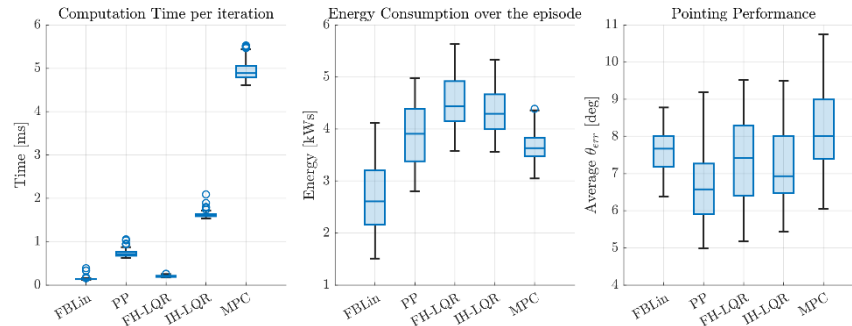


Figure 1. Simulation results for all controllers, using a matching MEKF model.

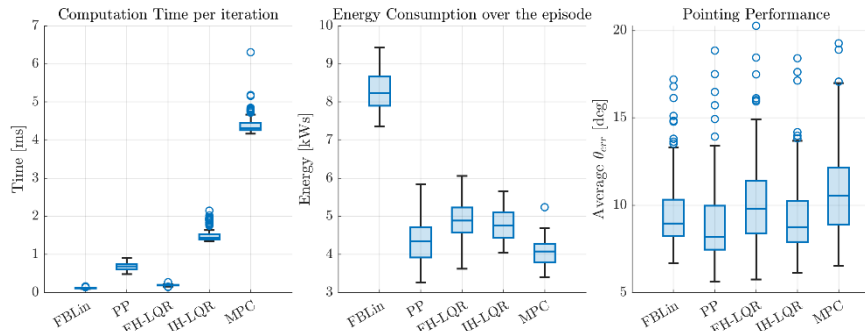


Figure 2. Simulation results for all controllers, using a badly tuned MEKF.

Third, only the MPC controller has been given the ability to provide a control law for the MTQs, which are the main source of power consumption. Designing other controllers using the extended state may reveal interesting and more power-efficient strategies regarding MTQ usage.

6. Conclusions

In conclusion, a thorough ADCS architecture has been presented, tailored to fit typical 3-Units CubeSat missions. Importance of S&A has been brought up, indicating redundancy needs for failure-tolerant behavior. It was followed by a comparative study of different control approaches, emphasizing robustness to model mismatch, and provided insight on low-consumption, high performance controllers. Among them, the finite-horizon linear quadratic regulator stands out, especially regarding its significantly lower computation effort. Future work may explore different control architectures and implement features regarding what has been discussed in Section 5.

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